

PML-AI Formula Sheet

PCI AI Project Management Leader — the complete quantitative reference

PERT · CPM · EMV · $E[\text{wait}]$ · WSJF · PTA

How to use this sheet

Every formula the PML-AI Body of Knowledge teaches, in one place, with the domain where it is worked in full.

Three columns matter. **Formula** is the expression. **Meaning** is what it answers. **Watch for** is the mistake that formula attracts in practice.

This sheet is a study aid. The examination specification, including whether any formula reference is provided, is confirmed by the Institute for each published examination form.

Conventions. Durations in working periods unless stated. Ratios and indices to 2 dp. Currency in USD unless stated. Where a formula is shared with the project-controls discipline, the PCL-AI sheet is the deeper treatment and is cross-referenced.

The delivery-approach rule. Several formulas below behave differently under predictive and adaptive delivery. Velocity forecasting and earned value answer the same question — when will this finish and at what cost — through different instruments. Under hybrid delivery, state which instrument governs the commitment before reporting either.

1. Notation

SYMBOL	MEANING	UNIT
PV, EV, AC	Planned value, earned value, actual cost	currency
BAC, EAC, ETC, VAC	Budget, estimate at completion, to complete, variance at completion	currency
CPI, SPI	Cost and schedule performance indices	ratio
O, M, P	Optimistic, most likely, pessimistic	periods
tE, σ	PERT expected duration, standard deviation	periods
ES, EF, LS, LF	Early start, early finish, late start, late finish	date
TF, FF	Total float, free float	periods
n	Number of stakeholders, resources or periods	count
WIP	Work in progress	items
EMV	Expected monetary value	currency
PTA	Point of total assumption	currency
r, t	Discount rate per period, period index	ratio, count

2. Business case and benefits · Domains 2, 16

FORMULA	MEANING	WATCH FOR	DOMAIN
$NPV = \sum CF_t \div (1 + r)^t$	Net present value of the investment	Benefits counted gross of the cost to realise them	2.3
$ROI = (Benefit - Cost) \div Cost$	Return on investment	Period over which benefit is counted left unstated	2.3
$BCR = PV(benefits) \div PV(costs)$	Benefit-cost ratio	Discounting one side and not the other	2.3
$Payback = Years\ before\ recovery + (Unrecovered \div Cash\ flow\ that\ year)$	Simple payback	—	2.3
$IRR: NPV = 0$	Internal rate of return	Multiple roots where signs change repeatedly	2.3
$Benefit\ realised = Actual\ measure - Baseline\ measure$	Benefits tracking	No baseline captured before delivery	16.3
$Benefits\ shortfall = Forecast\ benefit - Realised\ benefit$	Realisation gap	Measured too early to be meaningful	16.3

The baseline rule. A benefit cannot be measured without a pre-delivery baseline. If the baseline was never captured, the realisation number is an assertion. Capture it before delivery starts, not at handover.

3. Estimating and scheduling · Domain 6

FORMULA	MEANING	WATCH FOR	DOMAIN
$tE = (0 + 4M + P) \div 6$	PERT expected duration (beta)	Three points from a single estimator	6.2
$tE = (0 + M + P) \div 3$	Triangular expected duration	Confused with the PERT weighting	6.2
$\sigma = (P - 0) \div 6$	PERT standard deviation	—	6.2
$\sigma^2\ path = \sum \sigma^2\ along\ the\ path$	Path variance	Deviations summed instead of variances	6.4
$EF = ES + Duration$	Forward pass	Off-by-one at period boundaries	6.3
$LS = LF - Duration$	Backward pass	—	6.3
$TF = LS - ES = LF - EF$	Total float	If the two forms disagree, the pass is wrong	6.3
$FF = \min(successor\ ES) - EF$	Free float	Lags on the successor link ignored	6.3
$Critical\ path: TF = 0$	Longest path through the network	Constraints creating artificial zero float	6.3
$Crash\ cost\ slope = (Crash\ cost - Normal\ cost) \div (Normal - Crash\ duration)$	Cost per period saved	Crashing off the critical path	6.5
$Analogous\ estimate = Past \times (this\ driver \div past\ driver)$	Top-down estimating	—	6.2
$Parametric\ estimate = Parameter \times Rate$	Rate-based estimating	Rate from a different scope basis	6.2

Compression, compared

METHOD	MECHANISM	COST	RISK ADDED
Crashing	Add resource to critical activities	Direct cost rises	Diminishing returns, coordination load
Fast-tracking	Overlap activities previously in sequence	Little direct cost	Rework risk rises materially

4. Cost and earned value · Domain 7

Shared with the project-controls discipline. The PCL-AI Formula Sheet §4 is the deeper treatment, including earned schedule.

FORMULA	MEANING	WATCH FOR	DOMAIN
$CV = EV - AC$	Cost variance	Negative is adverse	7.3
$SV = EV - PV$	Schedule variance in currency	Converges to zero at completion however late	7.3
$CPI = EV \div AC$	Cost efficiency	AC excluding accruals	7.3
$SPI = EV \div PV$	Schedule efficiency in cost terms	—	7.3
$EAC = AC + (BAC - EV)$	Variance was a one-off	—	7.4
$EAC = BAC \div CPI$	Performance to date persists	—	7.4
$EAC = AC + (BAC - EV) \div (CPI \times SPI)$	Cost and schedule pressure both persist	—	7.4
$VAC = BAC - EAC$	Variance at completion	—	7.4
$TCPI = (BAC - EV) \div (BAC - AC)$	Efficiency required to hit budget	Quoted without comparing to CPI achieved	7.4
$\text{Cost to date} = \text{Actuals} + \text{Accruals}$	True cost to date	The commonest omission in the discipline	7.2
$\text{Resource utilisation} = \frac{\text{Productive hours}}{\text{Available hours}}$	Utilisation	Target set at 100%, which leaves no capacity for variability	7.5

Which EAC. The formula is not the difficulty; the assumption is. Was the variance a closed one-off, does performance persist, or are cost and schedule compounding? State the assumption and publish a range rather than a point.

5. Risk and uncertainty · Domain 8

FORMULA	MEANING	WATCH FOR	DOMAIN
$EMV = Probability \times Impact$	Expected monetary value	Applied to a single binary event	8.2
$Total\ EMV = \sum (P_i \times I_i)$	Portfolio expected value	—	8.2
$Risk\ score = Probability\ rating \times Impact\ rating$	Qualitative ranking	Ordinal ratings multiplied as if they were cardinal	8.2
$Contingency \approx \sum EMV, \text{ or } P80 - P50 \text{ from simulation}$	Risk-based contingency	Percentage of budget used instead	8.3
$EV\ of\ a\ decision\ branch = \sum (P \times outcome)$	Decision tree	Ignoring the cost of the option itself	8.4
$Value\ of\ perfect\ information = EV(with\ information) - EV(without)$	Whether to buy analysis	—	8.4
$Exposure\ after\ response = Residual\ probability \times Residual\ impact$	Residual risk	Response cost not netted off	8.3
$P50 / P80$	Confidence levels from simulation	P80 quoted as if it were a maximum	8.3

Ordinal ratings do not multiply. A $3 \times 4 = 12$ heat-map score is a sorting device, not a quantity. It cannot be summed across risks, and it cannot size a contingency. Use expected value or simulation for anything that touches money.

6. Agile, adaptive and flow · Domain 13

FORMULA	MEANING	WATCH FOR	DOMAIN
$Velocity = Points\ completed \div Sprint$	Delivery rate	Best-ever used rather than rolling average	13.3
$Sprints\ remaining = Points\ remaining \div Average\ velocity$	Forecast to completion	Backlog growth ignored	13.3
$Capacity = Team\ members \times Available\ days \times Focus\ factor$	Sprint capacity	Focus factor set at 1.0	13.3
$Cycle\ time = WIP \div Throughput$	Little's Law	Only valid for a stable system over the window measured	13.4
$Throughput = Items\ completed \div Period$	Delivery rate in items	—	13.4
$Lead\ time = Cycle\ time + Queue\ time$	Customer-observed duration	Reported as cycle time	13.4
$Flow\ efficiency = Active\ time \div Total\ elapsed\ time$	Proportion of time actually worked	—	13.4
$WSJF = Cost\ of\ delay \div Job\ size$	Weighted shortest job first	Cost of delay guessed rather than derived	13.3
$\% \text{ complete} = Points\ completed \div Total\ planned\ points$	AgileEVM progress	Point inflation across sprints	13.5
$Run\ rate = Team\ cost \div Sprint$	Cost per sprint	—	13.5
$Burn-up: \text{scope line and completed line plotted together}$	Progress against a moving scope	Burn-down hides scope growth; burn-up does not	13.3

Why burn-up beats burn-down for governance. A burn-down chart shows work remaining, so scope added mid-flight looks like slow progress. A burn-up plots completed work and total scope as two lines, which makes scope growth visible as what it is. For any project reporting to a gate, use burn-up.

7. Procurement and contracts • Domain 10

FORMULA	MEANING	WATCH FOR	DOMAIN
$PTA = [(Ceiling\ price - Target\ price) \div Buyer\ share\ ratio] + Target\ cost$	Point of total assumption (FPIF)	Seller's share ratio used in the denominator	10.3
$Fee = Target\ fee + Share\ ratio \times (Target\ cost - Actual\ cost)$	Incentive fee	Sign convention on an overrun	10.3
$Final\ price = Actual\ cost + Fee$ (capped at ceiling)	FPIF settlement	Ceiling not applied	10.3
$Pain/gain = Share\ ratio \times (Actual - Target)$	Target-cost mechanism	—	10.3
$LD\ exposure = LD\ rate \times Days\ late$	Liquidated damages	Cap not applied	10.4
Make-or-buy: compare total cost of ownership, not price	Sourcing decision	Transition and exit costs excluded	10.2
$Weighted\ score = \Sigma (criterion\ weight \times score)$	Tender evaluation	Weights set after seeing the bids	10.2

What the PTA means. Above the point of total assumption, the seller absorbs every further dollar of cost — the buyer's contribution has reached the ceiling. It is the point at which the contract's incentive stops working and the seller's interest changes character. **Check:** at an actual cost exactly equal to the PTA, the computed final price equals the ceiling price precisely. If it does not, the share ratio has been applied the wrong way round.

8. Quality and assurance • Domain 9

FORMULA	MEANING	WATCH FOR	DOMAIN
$Cost\ of\ quality = Prevention + Appraisal + Internal\ failure + External\ failure$	Total quality cost	Only failure costs counted	9.2
$DPMO = (Defects \div (Units \times Opportunities)) \times 1,000,000$	Defects per million opportunities	Opportunity count defined inconsistently	9.3
$Defect\ density = Defects \div Size$	Quality rate	—	9.3
$Rework\ ratio = Rework\ effort \div Total\ effort$	Rework burden	—	9.3
$Control\ limits = Mean \pm 3\sigma$	Statistical process control	Control limits confused with specification limits	9.3
$Process\ capability = (USL - LSL) \div 6\sigma$	Capability index	—	9.3
$First-pass\ yield = Units\ passing\ first\ time \div Units\ started$	Right-first-time	—	9.3

Control limits are not specification limits. Control limits describe what the process does; specification limits describe what the customer requires. A process can be in control and incapable at the same time, and the response to each is different.

9. Stakeholders, teams and communication · Domains 11, 12

FORMULA	MEANING	WATCH FOR	DOMAIN
$\text{Channels} = n(n - 1) \div 2$	Communication channels among n people	Growth is quadratic, and this is the point	11.2
$\Delta \text{Channels} = n$ when one person joins a team of n	Marginal communication load	—	11.2
Influence/interest grid position	Stakeholder strategy	Assessed once and never revisited	11.1
RACI: exactly one A per activity	Accountability rule	More than one A means none	12.2
Team productivity $\neq \Sigma$ individual productivity	Coordination overhead	—	12.3

Worked check. A team of 9 has $9 \times 8 \div 2 = 36$ channels. Adding one person takes it to 45 — nine new channels for one new person. This is the arithmetic behind small-team preference, and it is why adding people to a late project rarely accelerates it.

10. Programmes and portfolios · Domain 15

FORMULA	MEANING	WATCH FOR	DOMAIN
Portfolio value = Σ NPV of component projects	Aggregate value	Interdependencies ignored	15.2
Scoring model = Σ (criterion weight $\times$ criterion score)	Prioritisation	Weights adjusted to justify a chosen answer	15.2
Capacity constraint: Σ resource demand $\leq$ Available capacity	Portfolio feasibility	Demand summed at 100% utilisation	15.3
Programme EV = Σ component EV	Aggregated performance	Components on different measurement bases	15.4
Benefit dependency: benefit realised only when all enablers delivered	Benefits logic	Partial delivery credited with full benefit	15.4

11. Worked micro-examples

Each is internally consistent and reproducible from this sheet.

PERT. $O = 8$, $M = 12$, $P = 22$.

- $tE = (8 + 48 + 22) \div 6 = 13.0$
- $\sigma = (22 - 8) \div 6 = 2.33 \cdot \sigma^2 = 5.44$
- Triangular for comparison = $(8 + 12 + 22) \div 3 = 14.0$ — a different method, a different answer

Float. $ES = 4$, duration 6, $LF = 14$.

- $EF = 10 \cdot LS = 8 \cdot TF = 8 - 4 = 4 = 14 - 10$ — the two forms agree

Point of total assumption. Target cost 100,000 · target price 113,000 (fee 13,000) · ceiling 120,000 · buyer share 80%.

- $PTA = [(120,000 - 113,000) \div 0.80] + 100,000 = 8,750 + 100,000 = 108,750$
- Check at $AC = 108,750$: overrun 8,750; seller absorbs 20% = 1,750; fee = 13,000 – 1,750 = 11,250; price = 108,750 + 11,250 = 120,000, exactly the ceiling price

Earned value. PV 168,000 · EV 151,200 · AC 170,400 · BAC 420,000.

- $CPI = 0.887$ · $SPI = 0.900$
- EAC range = **439,200** (one-off) to **473,300** (CPI persists) to **507,000** (both persist)
- $TCPI$ to $BAC = (420,000 - 151,200) \div (420,000 - 170,400) = 1.08$ against a CPI of 0.89 — the budget requires a 22% efficiency improvement on work that has so far run 11% below plan

Little's Law. $WIP = 12$ items, throughput = 4 items per week.

- Cycle time = $12 \div 4 = 3$ weeks. Halving WIP to 6 halves cycle time to 1.5 weeks at the same throughput.

Communication channels. $n = 9 \rightarrow 36$ channels. $n = 10 \rightarrow 45$.

12. The ten formulas most often misapplied

1. **Risk scores multiplied and summed.** Ordinal heat-map ratings are a sorting device. They cannot size a contingency.
2. **SV used as a schedule measure late in a project.** Denominated in currency; converges to zero at completion however late. Use earned schedule.
3. **PTA computed with the seller's share ratio.** The denominator is the **buyer's** share. Check that the price at PTA equals the ceiling.
4. **Velocity taken as best-ever.** Use a rolling average across recent, comparable sprints.
5. **Burn-down used for gate reporting.** It conceals scope growth. Burn-up shows both lines.
6. **Standard deviations summed along a path.** Variances add; take the square root at the end.
7. **Crashing an activity off the critical path.** Cost incurred, no duration saved.
8. **Control limits confused with specification limits.** In control and incapable are different states with different responses.
9. **Resource utilisation targeted at 100%.** Leaves no capacity to absorb variability; queues grow non-linearly as utilisation approaches full.
10. **Benefits claimed without a pre-delivery baseline.** Without the baseline the realisation figure cannot be computed, only asserted.

13. Cross-references

FOR THE FULL TREATMENT	SEE
Every formula worked with figures and exercises	The PML-AI Body of Knowledge, at the domain cited
Earned value, earned schedule and forecasting in depth	PCL-AI Formula Sheet, §4
Investment appraisal and cost of capital in depth	PFL-AI Formula Sheet, §3 and §4
Why forecasts fail to change decisions	Knowing Early, Knowledge Series 01, §5

Website — <https://projectcontrolsinstitute.org>

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